

Case Study #2: American Wage Discrimination against Black Men in 1988

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2024-11-18

Introduction

Wage discrimination against Black Americans is an enduring indicator of racial inequality in the American labor force. In 1988, 25,632 male respondents reported information on their weekly wages in the previous year, years of education, years of experience, race (indicating Black or white), proximity to a metropolitan area, and their region of country (categorized as Northeast [E], Midwest [MW], South [S], and West [W]) in the U.S. Census Bureau's monthly Current Population Survey (CPS). The U.S. Census Bureau suspects that the racial wage gap between Black and white male workers, as measured in weekly wages, is dependent on the region of the country they indicated. We will investigate this suspicion by fitting a multiple regression model using the explanatory variables listed above to estimate the response variable of weekly wages, with an interaction term between race and region of country. By evaluating the magnitude and statistical significance of this interaction, and by controlling for education and experience, we will answer how large the wage gap was between Black and white workers in each region of the country, whether the size of this wage gap differed significantly in different regions, and whether Black males were paid less than their white counterparts with similar education and experience.

Data Description and Exploratory Data Analysis

This case study uses data from Black and white male respondents from the CPS conducted in March of 1988, and contains 25632 observations. We will use the variables described in the introduction section, with the exception of the SMSA variable for our data analysis, with weekly wages measured as a continuous numerical variable, years of education and years of experience as ordinal numerical variables, race as a binary categorical variable, and with region of country as a categorical variable with four factors. To form a general understanding of our data and its patterns of distribution, we take a look at the summary statistics for each variable. The Standard Metropolitan Statistical Area (SMSA) variable is excluded, as it is not a relevant factor in our case study.

Table 1: Summary Statistics for Weekly Wages

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
50.39	356.13	567.23	640.1625	826.21	18777.2

Table 2: Summary Statistics for Years of Education

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	12	12	13.07627	16	18

Table 3: Summary Statistics for Years of Experience

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-4	9	16	18.58656	27	63

Table 4: Summary Statistics for Race

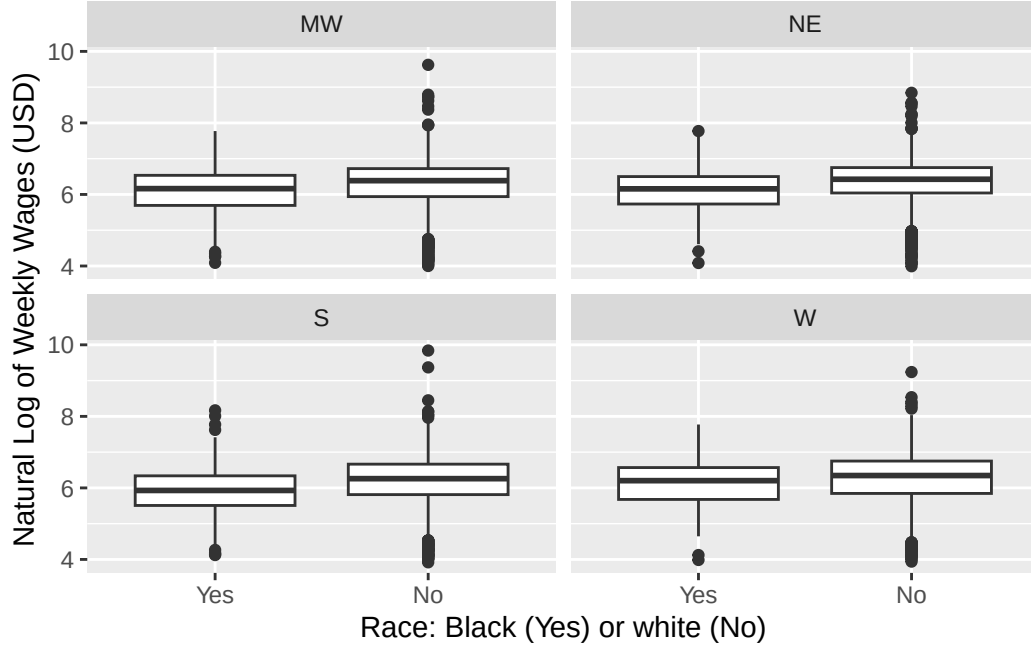
Black	White
1988	23643

Table 5: Summary Statistics for Region of Country

MW	NE	S	W
6226	5949	7991	5465

The observations for weekly wages range between 50.39 to 18,777.20 USD, with a mean of 640.16 USD, years of education ranged from 0 to 18 with a mean of 13.08 years, years of experience ranged from -4 to 63 with a mean of 18.59 years, the Black variable contains 1988 observations for Black men and 23643 observations for white men, and the Region variable contains 6226 observations in the Midwest, 5949 observations in the Northeast, 7991 observations in the South, and 5465 observations in the West. The years of experience variable is defined to be the individual's age - years of schooling - 6, so if an individual concluded 12 years of education prior at 16 years old, their years of experience will be a negative value ($16 - 12 - 6 = -2$). Aside from these negative numbers, there are no other unordinary or otherwise missing values in this data set.

Upon evaluating the box plots produced from our variables, we find that the box plots have extreme outliers that make it difficult to assess and compare the box plots to one another. We find that the conditions of linearity, equal variance, and normality conditions are violated for the model fit from our untransformed variables, so a natural log transformation was applied to the response variable Wage. The following box plot visualizes the comparison between the log value of weekly wages earned by Black (Yes) and white (No) men in each of the four regions recorded in the CPS.



From these sets of box plots, we find that there looks to be a consistently greater median natural log of weekly wages for white men than Black men in each region of the country. It is important to note that the number of Black individuals who are represented in this survey data is 1988 to the 23643 of white individuals represented. This disparity in standard deviations is expected, as standard deviations grow due with increasing sample size. However, from this data visualization alone, we cannot conclude any statistically significant differences between the weekly wages of Black and white men given different geographical regions.

Statistical Methods and Results

We fit a multiple regression to the variables to assess the statistical significance of the interaction between the Race and Region variables. We will proceed with a natural log transformation applied to the response Wage, as the log transformed model better satisfies conditions of linearity, equal variance, and normal residuals.

Our Full Model:

$$\begin{aligned}
 E[\log(Wage_i) | BlackNo_i, RegionNE_i, RegionS_i, RegionW_i, Education_i, Experience_i] = \\
 \beta_0 + \beta_1 BlackNo_i + \beta_2 RegionNE_i + \beta_3 RegionS_i + \\
 \beta_4 RegionW_i + \beta_5 Education_i + \beta_6 Experience_i \\
 + \beta_7 (BlackNo_i)(RegionNE_i) + \beta_8 (BlackNo_i)(RegionS_i) + \beta_9 (BlackNo_i)(RegionW_i)
 \end{aligned}$$

where $\log(Wage_i)$ denotes the natural logarithm of the i -th individual's weekly wage value measured in 1992 dollars, $BlackNo_i$ denotes whether the i -th individual is Black (0) or white (1), $RegionNE_i$ denotes whether the region the i -th individual is employed in is in the North East (1) or not (0), $RegionS_i$ denotes whether the region the i -th individual is employed in is in the South (1) or not (0), $RegionW_i$ denotes whether the region the i -th individual is employed

in is in the West (1) or not (0) , $Education_i$ denotes the number of years of schooling the $i - th$ individual received, and $Experience_i$ denotes the value of the $i - th$ individual's age minus the number of years of schooling minus an additional 6. The value of $RegionMW_i$, the effect of the $i - th$ individual being in the Midwest is inherent in the model as the reference level. The following table provides the estimates of the model parameters and their corresponding, relevant statistics.

Table 6: Model Summary Table

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.4673281	0.0345950	129.1323337	0.0000000
BlackNo	0.1928006	0.0306687	6.2865551	0.0000000
RegionNE	0.0571640	0.0420270	1.3601718	0.1737875
RegionS	-0.0965601	0.0337339	-2.8624030	0.0042078
RegionW	0.0415862	0.0504114	0.8249352	0.4094160
Education	0.0989226	0.0012075	81.9261048	0.0000000
Experience	0.0182709	0.0002811	64.9916204	0.0000000
BlackNo:RegionNE	0.0035277	0.0431955	0.0816678	0.9349115
BlackNo:RegionS	0.0417635	0.0350495	1.1915586	0.2334454
BlackNo:RegionW	-0.0382073	0.0514264	-0.7429495	0.4575190

As this model will give us the natural logarithm of the mean wage value, we will exponentiate the resulting value to interpret the 95% Confidence Interval for the pay differences by region. When we hold education and experience constant, we are 95% confident that the estimated weekly wage gap between white and Black men varied between 1.141896 USD and 1.287769 USD in the Midwest, varied between 1.137322 USD and 1.302103 USD in the Northeast, varied between 1.103004 USD and 1.449314 USD for the South, and varied between 1.027736 and 1.325551 USD for the West.

In order to answer whether region significantly modifies the relationship between race and wages, we perform a nested F test for the interaction coefficients β_7 , β_8 , and β_9 .

Hypothesis test in terms of coefficients:

$$H_0 : \beta_7 = \beta_8 = \beta_9 = 0 , H_A : \text{at least one of } \beta_7, \beta_8, \text{ and } \beta_9 \neq 0$$

Hypothesis test in terms of model comparisons:

$$H_0 : E[\log(Wage_i) | BlackNo_i, RegionNE_i, RegionS_i, RegionW_i, Education_i, Experience_i] = \beta_0 + \beta_1 BlackNo_i + \beta_2 RegionNE_i + \beta_3 RegionS_i + \beta_4 RegionW_i + \beta_5 Education_i + \beta_6 Experience_i$$

$$H_A : E[\log(Wage_i) | BlackNo_i, RegionNE_i, RegionS_i, RegionW_i, Education_i, Experience_i] = \beta_0 + \beta_1 BlackNo_i + \beta_2 RegionNE_i + \beta_3 RegionS_i + \beta_4 RegionW_i + \beta_5 Education_i + \beta_6 Experience_i + \beta_7 (BlackNo_i)(RegionNE_i) + \beta_8 (BlackNo_i)(RegionS_i) + \beta_9 (BlackNo_i)(RegionW_i)$$

Upon conducting the F-test, we fail to reject the null hypothesis and cannot conclude that region significantly modifies the relationship between race and wages ($p = 0.7837$). We cannot conclude that the size of this pay gap differs significantly in different regions.

On average, Black males were indeed paid less than their white counterparts with similar education and experience. This is indicated in the β_1 coefficient for the *BlackNo* variable, which is 0.1928006, a positive coefficient that indicates that if BlackNo is (1), that for every individual who is white, their estimated weekly wages are 1.212641 USD more than that of a Black individual of the same region, education, and experience. This is a statistically significant value as well, as we see the coefficient's p value is well below our alpha level 0.05, at $3.3e-10$, which is indicated as 0.0000000 in the Model Summary Table.

Discussion

The U.S. Census Bureau's suspicion that the extent to which Black workers were paid differently than white workers depending on the region of country cannot be confirmed, as the differences between the wage disparities across the four regions were found to be statistically insignificant ($p = 0.7837$). Nonetheless, there was a statistically significant difference between the weekly wages of Black workers compared to their white counterparts on average in every region of the country, holding education and experience constant ($p = 3.3e-10$). The 95% Confidence Interval calculated for all regions demonstrated how large the pay gap between Black and white workers was in each region of the country, which is summarized in the following: Midwest: [1.141896 USD, 1.287769 USD], Northeast: [1.137322 USD, 1.302103 USD], South: [1.103004 USD, 1.449314 USD], West: [1.027736 USD, 1.325551 USD]

The population that our results generalized to 23643 white workers and 1988 Black workers, with ranges of education and experience, across the four regions (Midwest, Northeast, South, and West), without accounting for metropolitan areas, nor age, specifically. One clear limitation of our analysis concerns this difference in number of observations for Black and white workers in this census data set. As there are nearly 12 times the number of white workers accounted for in the census to the number of Black workers in the census, the standard deviations across the race factor was indeed disproportionate. Perhaps the analysis would be more accurate and/or the differences between weekly wages between regions would be more statistically significant with nearly equal or similar numbers of observations for Black and white workers.

As we did not account for proximity to metropolitan areas in our analysis using the SMSA variable, we may also be failing to account for specifically wealthier vs. less wealthy areas. In the summary table of a model that includes SMSA as an explanatory variable to predict weekly wages, the p value associated with SMSANo (for non-metropolitan areas) was $2e-16$, a variable of strong statistical significance. Failing to control for cities/non-cities may have factored into less-informed conclusions about the relationship between race and wages across regions.

All of the explanatory variables included in the model indeed help to predict the weekly wages of Black and white workers based on their education and experience level as well as their U.S. region, but this relationship cannot be interpreted as a cause and effect relationship. This analysis is based on observational data that was collected in a census, wherein none of the factor levels were randomly assigned as they would be in an experiment. A data set of experimental data would allow us to conclude causal relationships, but our analysis and conclusions are merely observational, concerned with correlation instead.

R Appendix

```
#Loading the data and necessary packages
#| message: FALSE
#| warning: FALSE
# Loading necessary packages
library(tidyverse)
library(knitr)
library(dplyr)
library(Stat2Data)
library(broom)
library(Sleuth2)      # the package containing the data for the case study
library(kableExtra)   # for creating nicely formatted tables in Quarto

# Reading in and saving the data
wages <- Sleuth2::ex1029

#summary statistics tables
wages_stats <- summary(wages$Wage)
wages_df <- data.frame(t(as.vector(wages_stats)))
colnames(wages_df) <- names(wages_stats)
kable(wages_df, caption = "Summary Statistics for Weekly Wages", format = "html")
```

Table 7: Summary Statistics for Weekly Wages

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
50.39	356.13	567.23	640.1625	826.21	18777.2

```
edu_stats <- summary(wages$Education)
edu_df <- data.frame(t(as.vector(edu_stats)))
colnames(edu_df) <- names(edu_stats)
kable(edu_df, caption = "Summary Statistics for Years of Education", format =
  ↪ "html")
```

Table 8: Summary Statistics for Years of Education

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	12	12	13.07627	16	18

```
exp_stats <- summary(wages$Experience)
exp_df <- data.frame(t(as.vector(exp_stats)))
colnames(exp_df) <- names(exp_stats)
kable(exp_df, caption = "Summary Statistics for Years of Experience", format =
  ↪ "html")
```

Table 9: Summary Statistics for Years of Experience

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-4	9	16	18.58656	27	63

```

race_stats <- summary(wages$Black)
race_df <- data.frame(t(as.vector(race_stats)))
colnames(race_df) <- names(race_stats)

race_df <- race_df %>%
  rename(Black = Yes, White = No)
kable(race_df, caption = "Summary Statistics for Race", format = "html")

```

Table 10: Summary Statistics for Race

Black	White
1988	23643

```

reg_stats <- summary(wages$Region)
reg_df <- data.frame(t(as.vector(reg_stats)))
colnames(reg_df) <- names(reg_stats)
kable(reg_df, caption = "Summary Statistics for Region of Country", format =
  ↪ "html")

```

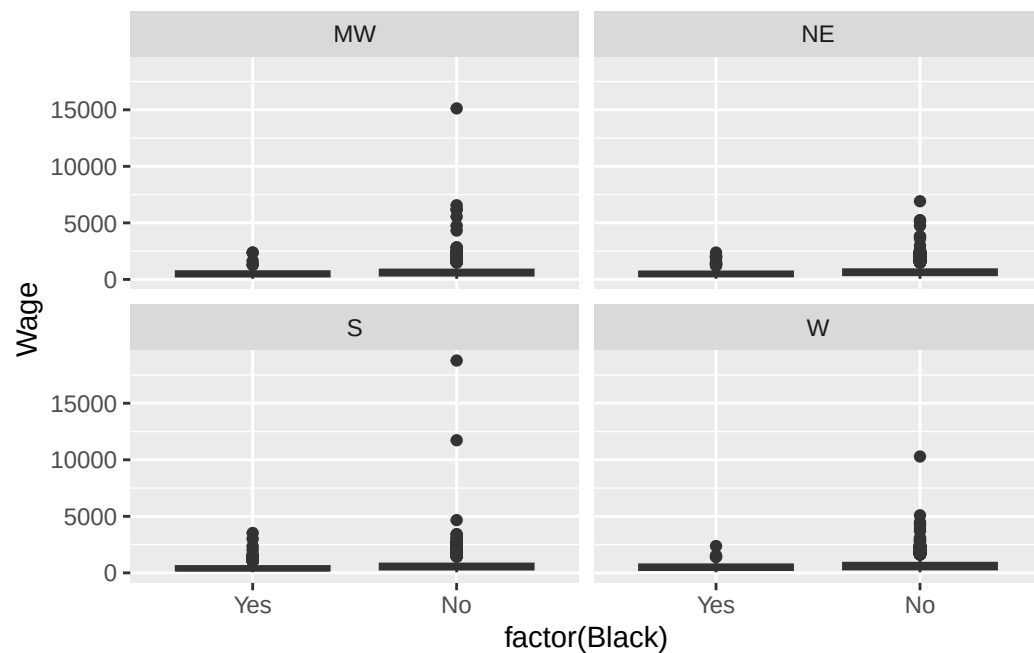
Table 11: Summary Statistics for Region of Country

MW	NE	S	W
6226	5949	7991	5465

```

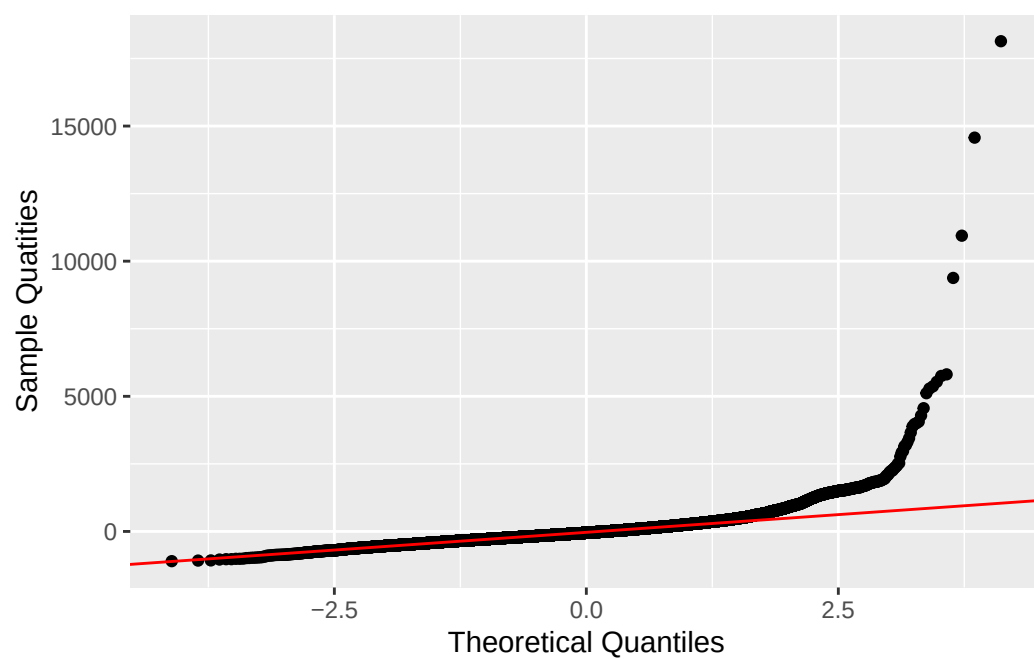
#Untransformed boxplots
qplot(x = factor(Black), y = Wage, data = wages, geom = "boxplot") +
  ↪ facet_wrap(vars(Region))

```

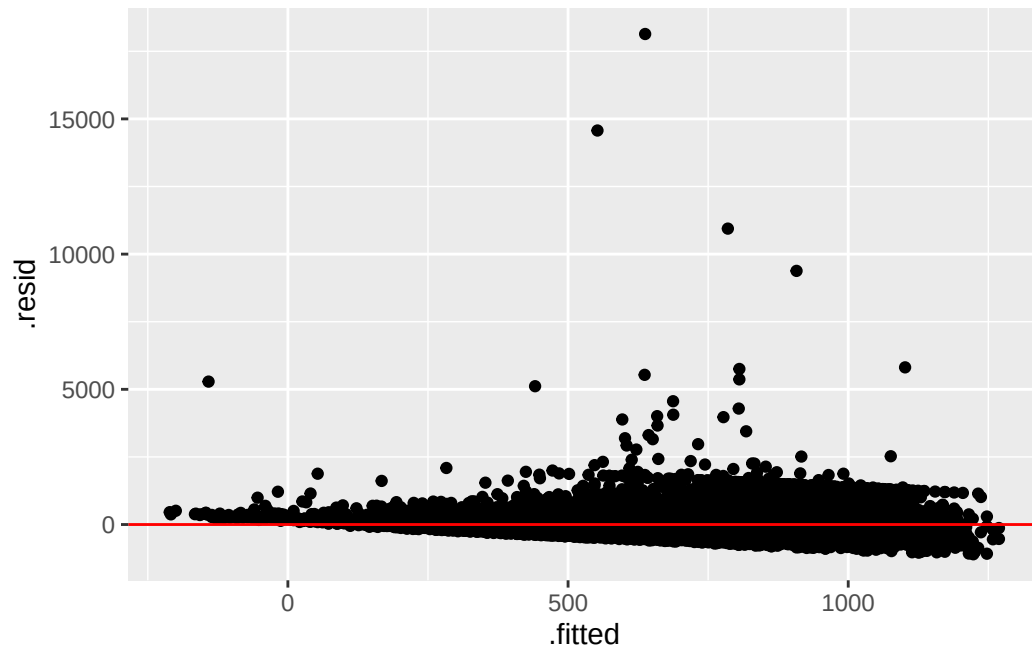


```
# Conditions Checking for untransformed model
wages_model <- lm(Wage ~ Black * Region + Education + Experience, data = wages)

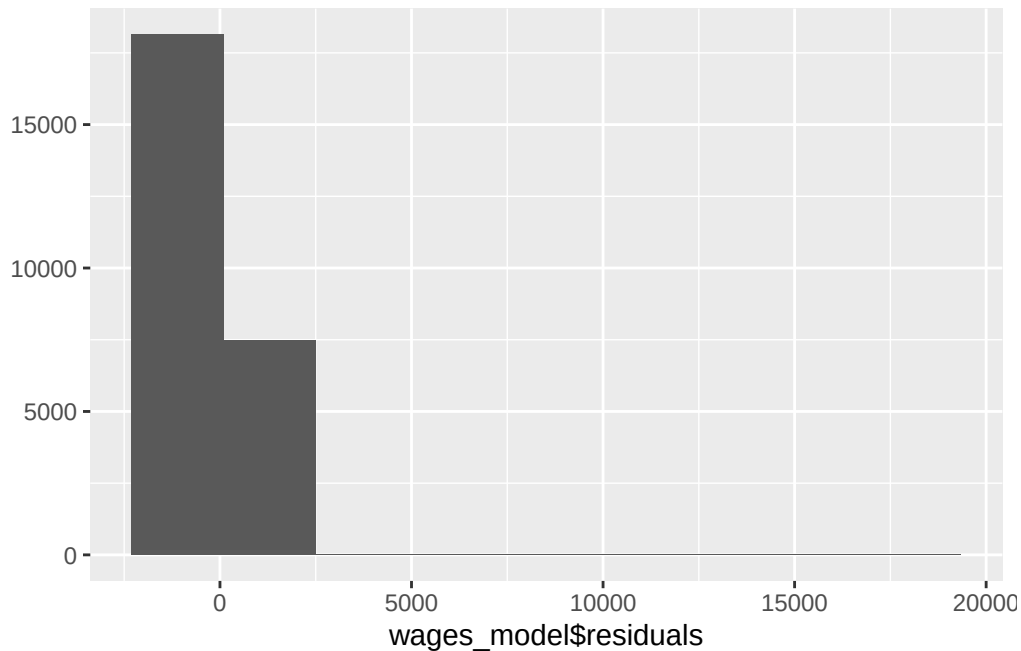
# Code to check Normality for the model (This is completely violated)
wages_model |>augment() |>ggplot(aes(sample = .resid)) + geom_qq()
  ↳ +geom_qq_line(col= "red") + xlab("Theoretical Quantiles") + ylab("Sample
  ↳ Quantities")
```



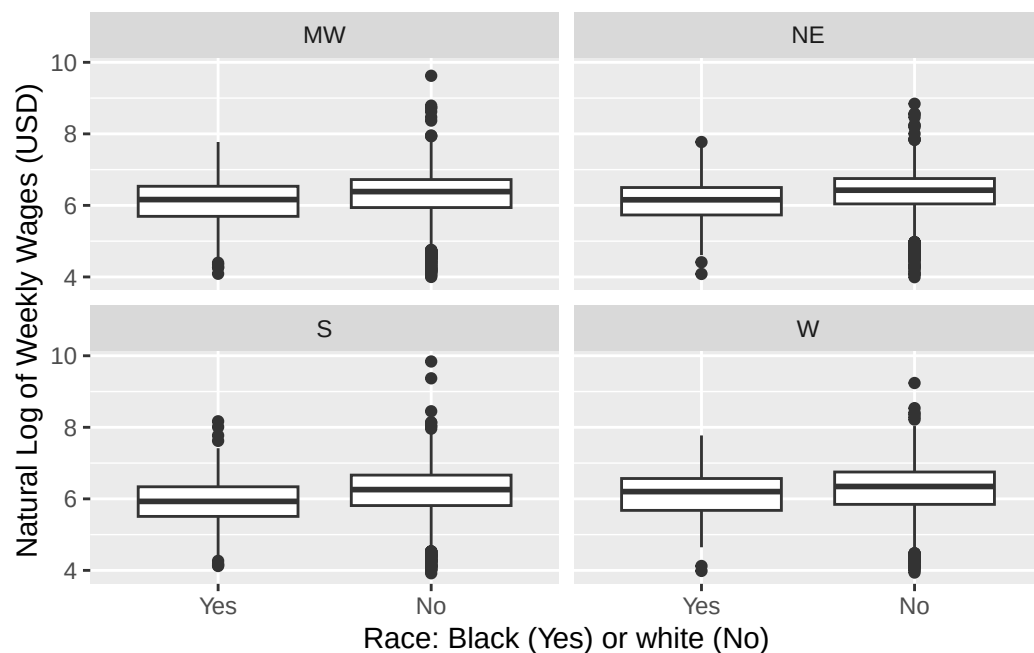
```
# Code to check linearity and equal variance for the wages model (This is
  ↳ completely violated for all the large y values, mostly)
wages_model |> augment() |> ggplot(aes(x = .fitted, y = .resid)) + geom_point() +
  ↳ geom_hline(yintercept = 0,col = "red")
```



```
#histogram for checking normality and skewness of residuals (This is completely
  ↳ violated, extreme right skewness)
qplot(x = wages_model$residuals, bins = 9)
```

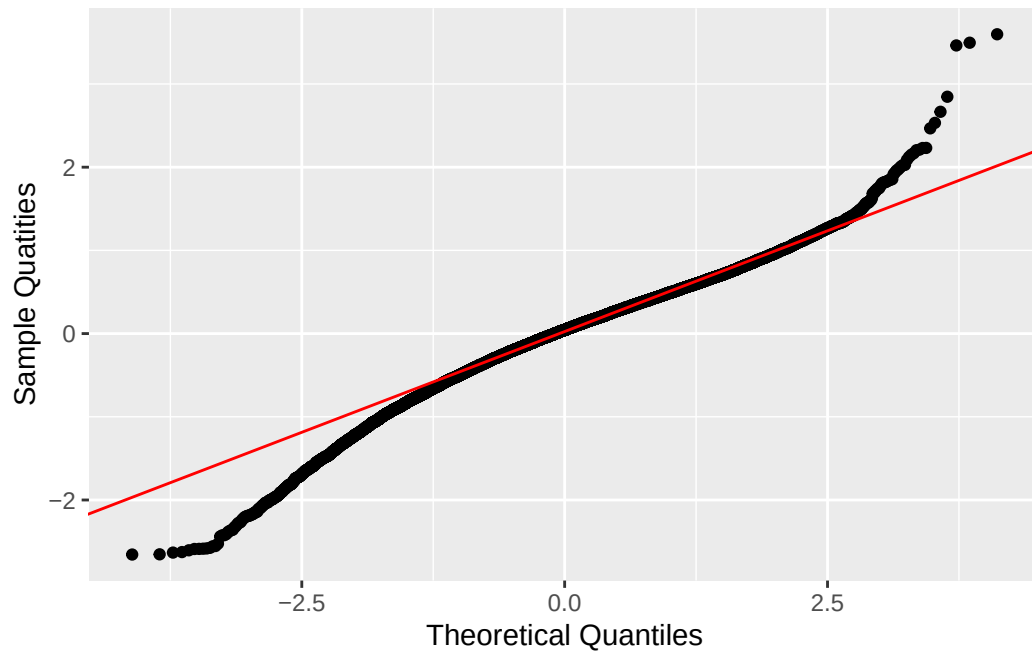


```
#Natural log transformed boxplots
qplot(x = factor(Black), xlab = "Race: Black (Yes) or white (No)", y = log(Wage),
  ↪ ylab = "Natural Log of Weekly Wages (USD)", data = wages, geom = "boxplot") +
  ↪ facet_wrap(vars(Region))
```

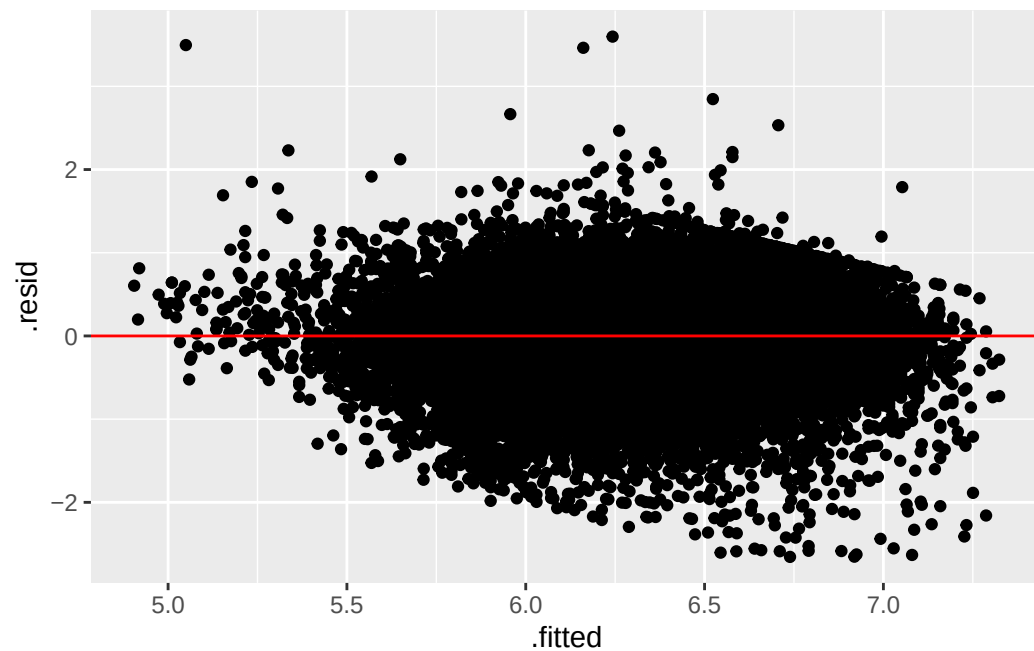


```
# Conditions Checking for natural log transformed model
wages_log_model <- lm(log(Wage) ~ Black * Region + Education + Experience, data =
  ↪ wages)
```

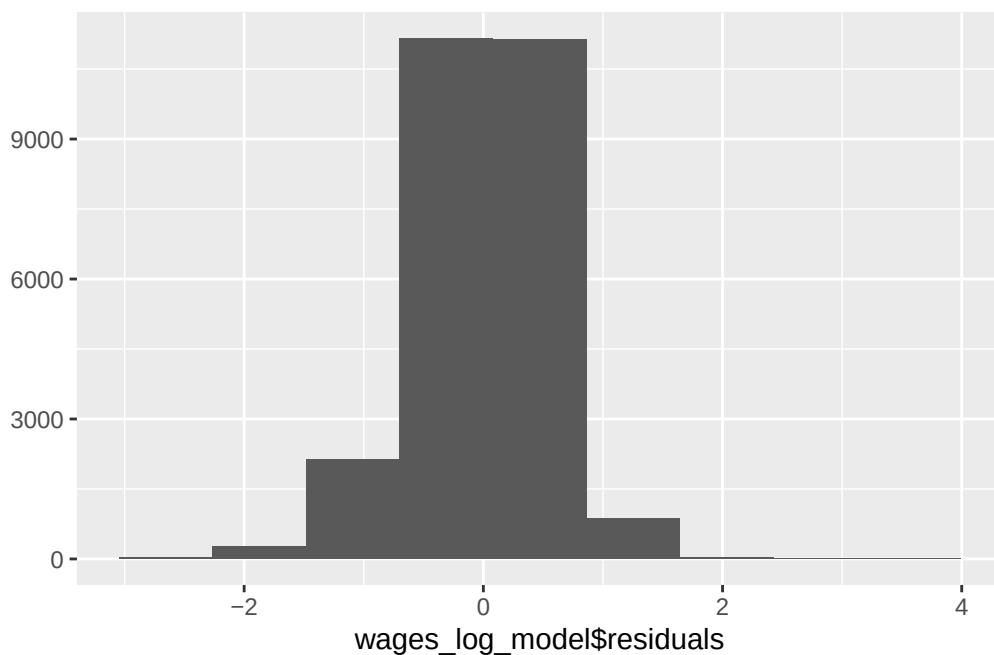
```
# Code to check Normality for the model (It's not perfect, but a significant
  ↳ improvement from the untransformed model)
wages_log_model |>augment() |>ggplot(aes(sample = .resid)) + geom_qq()
  ↳ +geom_qq_line(col= "red") + xlab("Theoretical Quantiles") + ylab("Sample
  ↳ Quatities")
```



```
# Code to check linearity and equal variance for the wages model (there is a bit
  ↳ of a cone shape from both sides where it widens in the middle, but because
  ↳ the residuals are of similar magnitude in comparison to the untransformed
  ↳ version, I consider this acceptable)
wages_log_model |> augment() |> ggplot(aes(x = .fitted, y = .resid)) +
  ↳ geom_point() + geom_hline(yintercept = 0,col = "red")
```



```
#histogram for checking normality and skewness of residuals (centered around 0,
↪ and no abundantly clear skewness)
qplot(x = wages_log_model$residuals, bins = 9)
```



```
#Summary tables of untransformed model
wages_model <- lm(Wage ~ Black * Region + Education + Experience, data = wages)
summary(wages_model)
```

```
Call:
lm(formula = Wage ~ Black * Region + Education + Experience,
    data = wages)
```

Residuals:

Min	1Q	Median	3Q	Max
-1099.7	-210.2	-49.8	143.5	18139.6

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-475.1390	25.6513	-18.523	< 2e-16 ***
BlackNo	104.6164	22.7401	4.601	4.23e-06 ***
RegionNE	25.8664	31.1619	0.830	0.4065
RegionS	-42.9668	25.0128	-1.718	0.0858 .
RegionW	20.5236	37.3788	0.549	0.5830
Education	62.2778	0.8953	69.561	< 2e-16 ***
Experience	10.7225	0.2084	51.440	< 2e-16 ***
BlackNo:RegionNE	10.0695	32.0283	0.314	0.7532
BlackNo:RegionS	22.4551	25.9883	0.864	0.3876
BlackNo:RegionW	-0.1473	38.1314	-0.004	0.9969

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 397.2 on 25621 degrees of freedom

Multiple R-squared: 0.201, Adjusted R-squared: 0.2007

F-statistic: 716.2 on 9 and 25621 DF, p-value: < 2.2e-16

```
#Summary table of natural log transformed model
log_wages_model <- lm(log(Wage) ~ Black * Region + Education + Experience, data =
  ↪ wages)
summary(log_wages_model)
```

Call:

```
lm(formula = log(Wage) ~ Black * Region + Education + Experience,
    data = wages)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.6555	-0.3022	0.0444	0.3511	3.5975

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.4673281	0.0345950	129.132	< 2e-16 ***
BlackNo	0.1928006	0.0306687	6.287	3.3e-10 ***
RegionNE	0.0571640	0.0420270	1.360	0.17379

```

RegionS      -0.0965601  0.0337339  -2.862  0.00421 **
RegionW      0.0415862  0.0504114   0.825  0.40942
Education    0.0989226  0.0012075  81.926  < 2e-16 ***
Experience    0.0182709  0.0002811  64.992  < 2e-16 ***
BlackNo:RegionNE 0.0035277  0.0431955   0.082  0.93491
BlackNo:RegionS 0.0417635  0.0350495   1.192  0.23345
BlackNo:RegionW -0.0382073  0.0514264  -0.743  0.45752

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5357 on 25621 degrees of freedom

Multiple R-squared: 0.2711, Adjusted R-squared: 0.2709

F-statistic: 1059 on 9 and 25621 DF, p-value: < 2.2e-16

```

#save to object for creating kable output
summary_table <- summary(log_wages_model)
coefficients_table <- as.data.frame(summary_table$coefficients)
kable(coefficients_table, caption = "Model Summary Table")

```

Table 12: Model Summary Table

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.4673281	0.0345950	129.1323337	0.0000000
BlackNo	0.1928006	0.0306687	6.2865551	0.0000000
RegionNE	0.0571640	0.0420270	1.3601718	0.1737875
RegionS	-0.0965601	0.0337339	-2.8624030	0.0042078
RegionW	0.0415862	0.0504114	0.8249352	0.4094160
Education	0.0989226	0.0012075	81.9261048	0.0000000
Experience	0.0182709	0.0002811	64.9916204	0.0000000
BlackNo:RegionNE	0.0035277	0.0431955	0.0816678	0.9349115
BlackNo:RegionS	0.0417635	0.0350495	1.1915586	0.2334454
BlackNo:RegionW	-0.0382073	0.0514264	-0.7429495	0.4575190

```

#process for constructing 95% confidence intervals for the difference in the
↪ weekly wages between Black and white individuals for each region

```

```

#begin with variance-covariance matrix for NE
vcov(log_wages_model)

```

	(Intercept)	BlackNo	RegionNE	RegionS
(Intercept)	1.196811e-03	-8.789310e-04	-8.921855e-04	-9.003074e-04
BlackNo	-8.789310e-04	9.405706e-04	8.908604e-04	8.908594e-04
RegionNE	-8.921855e-04	8.908604e-04	1.766271e-03	8.911498e-04
RegionS	-9.003074e-04	8.908594e-04	8.911498e-04	1.137978e-03
RegionW	-8.817795e-04	8.920024e-04	8.907401e-04	8.909721e-04

Education	-2.013315e-05	-1.007669e-06	2.418552e-07	5.649084e-07
Experience	-2.751647e-06	1.916885e-08	-1.030657e-07	1.060699e-07
BlackNo:RegionNE	8.946681e-04	-9.394412e-04	-1.766213e-03	-8.912381e-04
BlackNo:RegionS	8.943895e-04	-9.398193e-04	-8.910267e-04	-1.137823e-03
BlackNo:RegionW	8.763308e-04	-9.408561e-04	-8.906944e-04	-8.908150e-04
	RegionW	Education	Experience	BlackNo:RegionNE
(Intercept)	-8.817795e-04	-2.013315e-05	-2.751647e-06	8.946681e-04
BlackNo	8.920024e-04	-1.007669e-06	1.916885e-08	-9.394412e-04
RegionNE	8.907401e-04	2.418552e-07	-1.030657e-07	-1.766213e-03
RegionS	8.909721e-04	5.649084e-07	1.060699e-07	-8.912381e-04
RegionW	2.541311e-03	-9.177834e-07	1.075419e-07	-8.907775e-04
Education	-9.177834e-07	1.457962e-06	9.670206e-08	-3.553675e-07
Experience	1.075419e-07	9.670206e-08	7.903200e-08	4.861201e-08
BlackNo:RegionNE	-8.907775e-04	-3.553675e-07	4.861201e-08	1.865848e-03
BlackNo:RegionS	-8.913309e-04	-9.716555e-08	-1.027953e-07	9.397002e-04
BlackNo:RegionW	-2.541526e-03	1.297498e-06	-7.183522e-08	9.392991e-04
	BlackNo:RegionS	BlackNo:RegionW		
(Intercept)	8.943895e-04	8.763308e-04		
BlackNo	-9.398193e-04	-9.408561e-04		
RegionNE	-8.910267e-04	-8.906944e-04		
RegionS	-1.137823e-03	-8.908150e-04		
RegionW	-8.913309e-04	-2.541526e-03		
Education	-9.716555e-08	1.297498e-06		
Experience	-1.027953e-07	-7.183522e-08		
BlackNo:RegionNE	9.397002e-04	9.392991e-04		
BlackNo:RegionS	1.228465e-03	9.398954e-04		
BlackNo:RegionW	9.398954e-04	2.644679e-03		

```
#extracting the variance for the BlackNo term (beta_1)
var_b_1 <- vcov(log_wages_model)[1,1]
var_b_1
```

```
[1] 0.001196811
```

```
#extracting the variance for the Northeast interaction term
var_b_7 <- vcov(log_wages_model)[7,7]
var_b_7
```

```
[1] 7.9032e-08
```

```
#extracting the covariance between these terms
covar_b1_b7 <- vcov(log_wages_model)[1,7]
covar_b1_b7
```

```
[1] -2.751647e-06
```

```
#finding the standard error for (beta_1 + beta_7)
seNE <- sqrt(var_b_1 + var_b_7 + (2*covar_b1_b7))
#constructing the 95% Confidence interval
#point estimate is b_1+b_7, which is 0.1928006 + 0.0035277
point_estimate_NE <- 0.1928006 + 0.0035277
NE_ci_lb <- point_estimate_NE - (1.96*seNE)
NE_ci_ub <- point_estimate_NE + (1.96*seNE)
exp(NE_ci_lb)
```

```
[1] 1.137322
```

```
exp(NE_ci_ub)
```

```
[1] 1.302103
```

```
#-----
#variance-covariance matrix for S
#extracting the variance for the South interaction term
var_b_8 <- vcov(log_wages_model)[8,8]
var_b_8
```

```
[1] 0.001865848
```

```
#extracting the covariance between these terms
covar_b1_b8 <- vcov(log_wages_model)[1,8]
covar_b1_b8
```

```
[1] 0.0008946681
```

```
#finding the standard error for (beta_1 + beta_8)
seS <- sqrt(var_b_1 + var_b_8 + (2*covar_b1_b8))
#constructing the 95% Confidence interval
#point estimate is b_1+b_8, which is 0.1928006 + 0.0417635
point_estimate_S <- 0.1928006 + 0.0417635
S_ci_lb <- point_estimate_S - (1.96*seS)
S_ci_ub <- point_estimate_S + (1.96*seS)
exp(S_ci_lb)
```

```
[1] 1.103004
```

```
exp(S_ci_ub)
```

```
[1] 1.449314
```

```
#-----  
#variance-covariance matrix for W  
#extracting the variance for the West interaction term  
var_b_9 <- vcov(log_wages_model)[9,9]  
var_b_9
```

```
[1] 0.001228465
```

```
#extracting the covariance between these terms  
covar_b1_b9 <- vcov(log_wages_model)[1,9]  
covar_b1_b9
```

```
[1] 0.0008943895
```

```
#finding the standard error for (beta_1 + beta_9)  
seW <- sqrt(var_b_1 + var_b_9 + (2*covar_b1_b9))  
#constructing the 95% Confidence interval  
#point estimate is b_1+b_9, which is 0.1928006 - 0.0382073  
point_estimate_W <- 0.1928006 - 0.0382073  
W_ci_lb <- point_estimate_W - (1.96*seW)  
W_ci_ub <- point_estimate_W + (1.96*seW)  
exp(W_ci_lb)
```

```
[1] 1.027736
```

```
exp(W_ci_ub)
```

```
[1] 1.325551
```

```
#confidence interval for Midwest  
point_estimate_MW <- 0.1928006  
seMW <- 0.0306687  
MW_ci_lb <- point_estimate_MW - (1.96*seMW)  
MW_ci_ub <- point_estimate_MW + (1.96*seMW)  
exp(MW_ci_lb)
```

```
[1] 1.141896
```

```
exp(MW_ci_ub)
```

```
[1] 1.287769
```

```
#conducting nested F tests in R using anova(reduced, full):  
# Fitting the full and reduced models  
interact_lm <- lm(Wage ~ Black * Region + Education + Experience, data = wages)  
  
parallel_lm <- update(interact_lm, . ~ . - Black:Region)  
  
# Conducting the nested F test  
anova(parallel_lm, interact_lm)
```

Analysis of Variance Table

Model 1: Wage ~ Black + Region + Education + Experience

Model 2: Wage ~ Black * Region + Education + Experience

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	25624	4042236918				
2	25621	4042067674	3	169244	0.3576	0.7837